

Review of «PROBABILISTIC SIAMESE NETWORK FOR LEARNING REPRESENTATIONS»

Nikita Okhotnikov

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Introduction: Learning Representations

- ▶ **Goal:** Train deep neural networks to produce meaningful vector representations (embeddings) of input data (e.g., images).
- ▶ This is often done using **weakly labelled information**: binary similarity labels for pairs of training images (similar/dissimilar).
- ▶ **Problem with Traditional Approaches:**
 - ▶ Use of **fixed, hand-crafted cost functions** (e.g., L1, L2 norms, mutual information).
 - ▶ These cost functions are often **not normalized**, leading to:
 - ▶ Potential biases in learned representations.
 - ▶ Difficulty in comparing similarity scores.
- ▶ **Paper's Core Idea:** Formulate learning as maximizing the likelihood of binary similarity labels under a **parameterized probabilistic similarity model**.

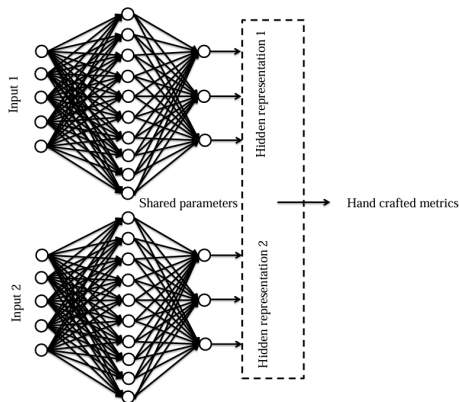
Standard Siamese Networks

Architecture:

- ▶ Two (or more) **identical sub-networks** with **shared weights** (θ).
- ▶ Each sub-network processes one input from a pair.

Process:

- ▶ Input: A pair of samples (X_1, X_2).
- ▶ Each sub-network $f_{\theta}(\cdot)$ maps its input to an embedding:
 $h_1 = f_{\theta}(X_1)$, $h_2 = f_{\theta}(X_2)$



Traditional Cost Functions (e.g., Contrastive Loss)

Let $Y \in \{0, 1\}$ be the label (1 if similar, 0 if dissimilar). $D_W = \|h_1 - h_2\|_2$.

- ▶ **For similar pairs** ($Y = 1$): Minimize distance. Cost: $E_{sim} = D_W^2$
- ▶ **For dissimilar pairs** ($Y = 0$): Push distance beyond a margin m . Cost: $E_{dissim} = \max(0, m - D_W)^2$

The Probabilistic Siamese Network (PSN) Approach

Key Innovation: Replace fixed cost functions with a learnable, probabilistic model of similarity.

- ▶ Given embeddings h_1, h_2 from the Siamese network.
- ▶ Define a model for the probability of the binary similarity label $s \in \{0, 1\}$:

$$P(s = 1|h_1, h_2, \alpha) = g(h_1, h_2, \alpha)$$

$$P(s = 0|h_1, h_2, \alpha) = 1 - g(h_1, h_2, \alpha)$$

where:

- ▶ $g(\cdot)$ is the **similarity function**, outputting a value in $[0, 1]$.
- ▶ α are **learnable parameters** of this similarity model.
- ▶ **Properties of $g(h_1, h_2, \alpha)$:**
 - ▶ Symmetric in h_1, h_2 : $g(h_1, h_2, \alpha) = g(h_2, h_1, \alpha)$.
 - ▶ Maximum similarity should occur when $h_1 = h_2$.
- ▶ The function g typically depends on a **weighted distance metric d** between h_1 and h_2 .

Mathematical Forms of the Similarity Model g

Function g often depends on a weighted norm d of the difference between h_1, h_2 :

$$d = \sum_{n=1}^N \alpha_n |h_{1,n} - h_{2,n}|^l$$

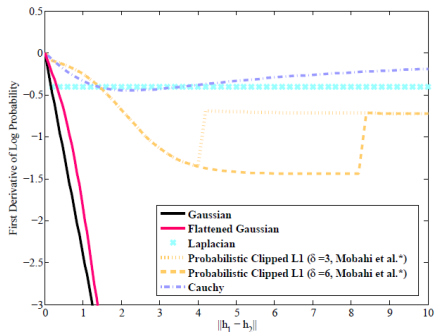
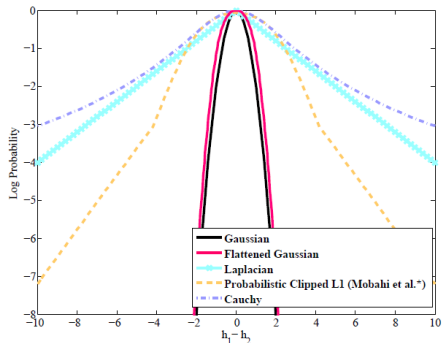
where N is the dimensionality of embeddings, $\alpha_n > 0$ are learnable, and $l = \overline{1, 2}$

Several Forms of $g(d)$ Explored:

- ▶ **Gaussian-like:** $g(d) = \exp(-d)$
 - ▶ where $d = \sum \alpha_n (h_{1,n} - h_{2,n})^2$.
 - ▶ Log P decreases quadratically for large d .
- ▶ **Laplacian-like:** $g(d) = \exp(-d)$, $d = \sum \alpha_n |h_{1,n} - h_{2,n}|$
 - ▶ Log P decreases linearly with d .
- ▶ **Cauchy-like (Best Performer in Paper):**
 $g(d) = \frac{1}{1+d}$, $d = \sum \alpha_n (h_{1,n} - h_{2,n})^2$
 - ▶ Log P decreases as $1/d^2$ (slower than Gaussian for large d).
 - ▶ More robust to outliers due to «heavy tails».
- ▶ **Probabilistic Clipped L1:** A normalized version of the clipped L1 cost.

$$g(d) = \frac{e^{-d}}{e^{-d} + e^{-\max(0, \delta-d)}} \quad \text{where } d = \sum \alpha_n |h_{1,n} - h_{2,n}|$$

Cost function comparison



Learning the Probabilistic Model

Objective: Maximize the conditional log-likelihood of the observed similarity labels $s^{(m)}$ for M training pairs.

$$\begin{aligned}\mathcal{L}(\theta, \alpha) &= \sum_{m=1}^M \log P(s^{(m)} | h_1^{(m)}, h_2^{(m)}, \alpha) \\ &= \sum_{m=1}^M \left[s^{(m)} \log g(h_1^{(m)}, h_2^{(m)}, \alpha) \right. \\ &\quad \left. + (1 - s^{(m)}) \log(1 - g(h_1^{(m)}, h_2^{(m)}, \alpha)) \right]\end{aligned}$$

where:

- ▶ $h_1^{(m)} = f_{\theta}(X_1^{(m)})$ and $h_2^{(m)} = f_{\theta}(X_2^{(m)})$.
- ▶ θ are the parameters of the Siamese sub-networks.
- ▶ α are the parameters of the similarity function g .

Key Advantages & Experimental Highlights

Advantages of the Probabilistic Approach

- ▶ **Principled Framework:** Eliminates the need for hand-crafting cost functions.
- ▶ **Natural Normalization:** Probabilities $P(s|h_1, h_2)$ are normalized ($[0, 1]$).
- ▶ **Adaptability:** The similarity model $g(\cdot, \alpha)$ itself is learned.
- ▶ **Robustness:** Certain forms (e.g., Cauchy-like) can better handle outliers and FP/NP.
- ▶ **Improved Interpretability** of similarity scores.

Experimental Highlights

- ▶ **Datasets:** MNIST, AT&T ORL faces, COIL-100 objects
- ▶ The **Cauchy-like similarity model** consistently performed well, outperforming Gaussian and Laplacian models.
 - ▶ AT&T Faces: Cauchy (94.30%) vs Clipped L1 (91.80%).
- ▶ **Competitive Results with Weak Labels:**
 - ▶ MNIST: 1.93% error (with PCA preprocessing), comparable to some deep networks trained on full class labels.
 - ▶ COIL-100: 70.67% accuracy, comparable to standard CNNs using class labels.

Conclusion

- ▶ The paper successfully introduces a probabilistic framework for training Siamese networks.
- ▶ The Cauchy-like similarity model seems to be superior due to its robustness.
- ▶ The learned representations are shown to be effective for downstream tasks
- ▶ Approach requires quite some data to train on
- ▶ Evaluation is questionable (MNIST?.. Two layer MLP?..)
- ▶ Non-binary similarity functions?? (Discussed as future work, but that seems so simple to implement..)